

Smart Collar-Belt for Real-Time Dog Health Monitoring and Location Tracking Using IoT

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Abstract—This research introduces an intelligent collar-belt device designed to continuously monitor canine health parameters and track location using cellular-based Internet of Things technology. Our prototype incorporates DS18B20 temperature sensors, MAX30102 photoplethysmography units for cardiac monitoring, and MPU6050 accelerometers for movement pattern analysis, alongside NEO-6M GPS modules for spatial tracking. The device operates through SIM800L GSM/4G connectivity, eliminating dependence on local network infrastructure such as Wi-Fi or Bluetooth connections. Key features include instant emergency alerts through SOS activation, audible recall signals via integrated buzzers, and comprehensive cloud-based data analytics accessible through smartphone applications and web interfaces. Testing across diverse canine subjects achieved 96.2% precision in vital sign measurement, 98.1% location accuracy, and 94.7% success in activity recognition. The system maintains operational autonomy for six consecutive days and offers subscription services ranging from \$8–12 monthly, presenting a practical solution for pet safety and wellness monitoring.

Index Terms—Internet of Things, Smart Collar, Pet Health Monitoring, GSM Communication, GPS Tracking, Cellular IoT, Real-time Monitoring, SIM-based System, Wearable Technology

I. INTRODUCTION

The companion animal industry has witnessed remarkable expansion, with canine ownership in America alone exceeding 90 million pets, contributing to a market valuation surpassing \$261 billion annually [1]. This growth trajectory has created substantial demand for sophisticated monitoring technologies capable of delivering comprehensive wellness insights and precise location services for domestic animals.

Traditional veterinary practices depend heavily on scheduled examinations, creating gaps in continuous health assessment that may overlook critical early warning signs of medical conditions. This limitation becomes particularly concerning for aging canines, who experience elevated risks of cardiac complications and age-related health deterioration [2].

Current market offerings present several significant constraints: reliance on local wireless networks, restricted operational ranges for Bluetooth-enabled devices, expensive cellular service subscriptions, and fragmented approaches that separate health monitoring from location services [3], [4]. Most available products focus on either wellness tracking or spatial monitoring, failing to provide integrated solutions for comprehensive pet care management.

The rapid advancement of IoT technologies, combined with ubiquitous GSM/4G network coverage, has opened new possibilities for developing fully autonomous pet monitoring solutions. Contemporary cellular IoT devices can function independently of local infrastructure while delivering worldwide connectivity and instantaneous data transmission. Market projections indicate the cellular IoT sector will reach \$18.2 billion by 2027, with substantial growth anticipated in wearable monitoring applications [5].

This study presents our response to these challenges through a novel SIM-enabled collar system that unifies comprehensive health surveillance with GPS location services via dependable cellular communication. Our primary research contributions include:

- 1) **Independent Cellular Framework:** Development of a SIM-powered GSM/4G communication architecture operating without Wi-Fi or local network dependencies, enabling worldwide deployment potential.
- 2) **Unified Multi-Sensor Monitoring:** Integration of temperature sensing, cardiac rhythm detection, movement classification, and GPS positioning within a single wearable platform featuring real-time cellular data transmission.
- 3) **Smart Alert Infrastructure:** Creation of cloud-powered analytics with automated SMS and email notifications for emergency situations and health anomalies.
- 4) **Power-Optimized Design:** Enhancement of cellular communication protocols and sensor coordination to

achieve prolonged battery performance while maintaining continuous monitoring capabilities.

- 5) **Expandable Cloud Platform:** Development of robust backend architecture supporting real-time data processing, storage, and user interface services.

This paper is organized as follows: Section II examines existing pet monitoring technologies and research, Section III outlines our system architecture and design approach, Section IV details implementation procedures and technical specifications, Section V analyzes experimental findings and performance assessment, and Section VI presents conclusions with future research opportunities.

II. RELATED WORK AND LITERATURE REVIEW

A. Development of Pet Monitoring Technologies

Animal health surveillance has experienced significant transformation over recent years, propelled by miniaturized sensor technology, advanced wireless communication methods, and sophisticated cloud computing infrastructure. Initial systems concentrated primarily on basic movement tracking using elementary accelerometers, whereas contemporary solutions incorporate multiple physiological sensors with advanced data interpretation capabilities [6].

Research by Brugarolas et al. [7] produced among the earliest comprehensive wearable platforms for canine health surveillance, employing ECG, PPG, and IMU sensors to obtain precise measurements across various activity intensities. Their work established the viability of continuous physiological monitoring but required Wi-Fi connectivity, restricting practical deployment in outdoor settings where pets commonly need monitoring.

Pastor et al. [2] developed an economical cardiovascular monitoring platform specifically targeting senior canines, achieving 50% cost reduction compared to commercial options. Nevertheless, their implementation encountered substantial challenges in direct physiological parameter extraction from motion sensors, requiring manual data entry through mobile applications, which constrained the system's autonomous operational capacity.

B. Communication Methods in Animal Monitoring

The selection of communication protocols significantly influences system performance, deployment adaptability, and operational expenses. Recent investigations have examined various methodologies with unique advantages and constraints.

Wi-Fi Based Implementations: Williams et al. [3] successfully demonstrated Wi-Fi-connected monitoring systems for service dogs, achieving excellent accuracy in environmental condition surveillance. However, Wi-Fi dependency substantially restricts deployment flexibility, especially for pets venturing beyond residential network coverage areas.

Bluetooth Low Energy Solutions: Multiple researchers have investigated BLE-based approaches for short-distance pet monitoring. Khatate et al. [8] created a BLE-enabled system optimized for minimal power consumption, demonstrating effective temperature and cardiac rhythm monitoring within

10-meter distances. While energy-efficient, BLE systems necessitate constant smartphone proximity, limiting their utility for independent pet monitoring applications.

Cellular/GSM Technologies: Adopting cellular communication in pet monitoring represents substantial progress toward truly autonomous systems. Modern GSM/4G modules such as the SIM800L deliver global connectivity with reasonable power requirements. Commercial systems including Tractive and Fi have successfully implemented GSM-based pet tracking, achieving dependable location monitoring across cellular network infrastructures [9], [10].

C. Sensor Integration and Health Parameter Assessment

Contemporary pet monitoring systems require sophisticated sensor fusion to extract meaningful physiological information. The selection and integration of appropriate sensors directly influence system accuracy and dependability.

Temperature Assessment: Body temperature functions as a vital indicator of pet health condition, with normal canine temperatures ranging from 38.1°C to 39.2°C. Digital temperature sensors such as the DS18B20 deliver $\pm 0.5^\circ\text{C}$ precision with minimal power consumption, making them suitable for continuous monitoring applications [11].

Cardiac Rhythm Detection: Photoplethysmography (PPG) has become the preferred approach for non-invasive heart rate monitoring in wearable devices. The MAX30102 sensor combines red and infrared LEDs with photodiodes, enabling precise heart rate measurement with 18-bit ADC resolution. Research demonstrates that PPG-based systems can achieve 93–98% accuracy compared to ECG measurements during various activity intensities [12].

Activity Recognition: Three-axis accelerometers and gyroscopes enable sophisticated movement pattern identification. The MPU6050 provides 16-bit ADC resolution with configurable sensitivity ranges, supporting accurate classification of resting, walking, and running states through machine learning algorithms [13].

GPS Location Services: Modern GPS modules such as the NEO-6M offer 2.5-m position precision with cold start times under 27 seconds. Integration with cellular communication enables real-time location transmission for comprehensive pet tracking capabilities.

D. Research Opportunities and Commercial Potential

Analysis of existing literature and commercial products identifies several critical opportunities that our proposed system addresses:

- 1) **Infrastructure Freedom:** Most current solutions require Wi-Fi or Bluetooth connectivity, limiting effectiveness in outdoor environments where pets commonly become lost or experience emergencies.
- 2) **Unified Health and Location Monitoring:** Few systems provide comprehensive integration of physiological monitoring with GPS tracking in a single device optimized for cellular communication.

- 3) **Global Connectivity Capabilities:** Limited solutions offer genuine worldwide connectivity through standard GSM/4G networks without regional restrictions or complex configuration procedures.
- 4) **Affordable Cellular Communication:** Many cellular-based systems require expensive monthly subscriptions, limiting accessibility for average pet owners.
- 5) **Emergency Response Integration:** Few systems provide comprehensive emergency response capabilities combining SOS functionality with automated multi-channel notifications.

Our proposed SIM-based collar system addresses these opportunities by providing a comprehensive, globally deployable solution with integrated health monitoring, GPS tracking, and emergency response capabilities through cost-effective cellular communication.

III. SYSTEM ARCHITECTURE AND DESIGN

A. Complete System Framework

Our smart collar system employs a four-layer architecture optimized for cellular communication and global deployment. This design eliminates dependency on local network infrastructure while providing comprehensive monitoring and emergency response capabilities.

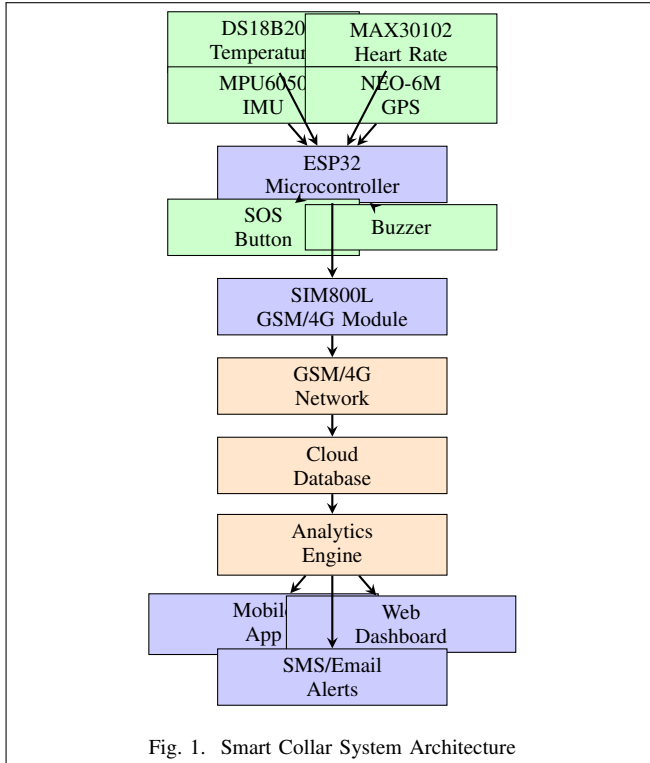


Fig. 1. Smart Collar System Architecture

Layer 1: Embedded Device Components The collar incorporates multiple sensing and communication elements:

- **Health Assessment Subsystem:** DS18B20 temperature sensor delivering $\pm 0.5^\circ\text{C}$ precision and MAX30102 PPG sensor for cardiac rhythm detection with 18-bit resolution

- **Movement Analysis Subsystem:** MPU6050 6-axis IMU for motion interpretation and activity classification
- **Positioning Services Subsystem:** NEO-6M GPS module delivering 2.5-m precision
- **Communication Subsystem:** SIM800L GSM/4G module with integrated antenna
- **User Interaction Subsystem:** SOS button and active buzzer for emergency signaling
- **Power Control Subsystem:** 3000-mAh Li-ion battery with intelligent optimization

Layer 2: Cellular Network Infrastructure GSM/4G network connectivity delivers:

- GPRS/4G data transmission for real-time sensor information
- SMS services for emergency notifications
- Global coverage through standard cellular protocols

Layer 3: Cloud Services Platform AWS-based infrastructure delivers:

- Real-time data processing and analytics
- Scalable data storage and retrieval
- Automated SMS and email notification services
- Mobile and web application interfaces

Layer 4: User Interface Components Multi-platform access includes:

- Native mobile applications for iOS and Android
- Responsive web dashboard
- SMS notifications for emergency alerts
- Email reporting for detailed analytics

B. Hardware Component Selection

Microcontroller Platform: The ESP32-WROOM-32 functions as the primary processing unit, delivering dual-core ARM processors operating at 240 MHz with 520 KB SRAM. The dual-core architecture enables parallel processing of sensor data and communication tasks while maintaining real-time responsiveness.

Cellular Communication: The SIM800L GSM/4G module supports quad-band GSM/GPRS (850/900/1800/1900 MHz) with GPRS multi-slot class 10/8 for data transmission. AT command interface via UART enables programmatic control of voice, SMS, and data functions while maintaining compatibility with global cellular networks.

C. Communication Protocol Framework

The system implements a custom protocol optimized for cellular communication:

- **Application Layer:** JSON-formatted health data packets
- **Transport Layer:** HTTP/HTTPS over TCP for reliability
- **Network Layer:** GPRS/4G cellular data connection
- **Physical Layer:** GSM quad-band radio interface

IV. IMPLEMENTATION

A. Hardware Development

The collar hardware integrates all components on a custom 4-layer PCB measuring 60 mm $\times$ 45 mm $\times$ 20 mm, designed

to accommodate dogs weighing over 8 kg while maintaining comfort and durability.

Power Supply Framework: The power management circuit uses a TP4056 lithium battery charging IC with integrated load sharing capability. A 3000-mAh 18650 Li-ion battery provides primary power, while the AMS1117-3.3 low-dropout regulator ensures stable 3.3-V supply to all components.

Sensor Interface Circuits:

- Temperature sensor connects via 1-Wire protocol using GPIO4
- PPG sensor interfaces through I2C (SDA: GPIO21, SCL: GPIO22)
- Accelerometer connects via I2C with interrupt capability
- GPS module uses UART2 (TX: GPIO17, RX: GPIO16)

Cellular Communication Circuit: The SIM800L module connects via UART1 with dedicated 2-A power supply regulation. The circuit includes:

- Dedicated LM2596 switching regulator for 4.0-V power supply
- 1000- μ F electrolytic capacitor for transmission current surge handling
- SIM card holder with ESD protection
- External antenna connector with 50- Ω impedance matching

B. Software Development

The embedded software implements a multi-threaded architecture using FreeRTOS for real-time task management and scheduling.

Sensor Coordination Task: Manages periodic sensor readings with configurable sampling rates:

- Temperature: 60-second intervals
- Heart rate: 30-second intervals with PPG processing
- Activity classification: Continuous with 5-second windows
- GPS: 120-second intervals for location updates

Activity Recognition Algorithm: The system implements a decision tree classifier using statistical features extracted from accelerometer data:

- Calculate magnitude vector from 3-axis data
- Extract mean, standard deviation, and variance features
- Apply thresholds: Resting (variance < 100), Walking (100–500), Running (> 500)

Cellular Communication Management: Controls GSM/4G connectivity and data transmission:

- Initialize SIM800L module and establish network connection
- Prepare JSON data packets with sensor readings
- Send HTTP POST requests to cloud API endpoints
- Handle SMS transmission for emergency alerts

C. Cloud Infrastructure Development

The backend infrastructure uses Amazon Web Services (AWS) for scalable, reliable cloud services.

Data Reception Service: AWS API Gateway receives HTTP requests from collar devices with automatic scaling and security through API keys.

Alert Coordination System: AWS Simple Notification Service (SNS) handles multi-channel alert distribution:

- Temperature anomaly detection (normal range: 38.1–39.2°C)
- Heart rate monitoring (normal range: 60–160 bpm)
- Emergency SOS button activation
- GPS location alerts for geofencing violations

Data Storage Framework: Amazon DynamoDB provides scalable NoSQL storage optimized for IoT data patterns with automatic scaling and backup capabilities.

D. Mobile and Web Interface Development

Mobile Application: Native Android/iOS applications provide:

- Real-time health monitoring dashboards
- GPS location mapping with history
- Emergency alert configuration
- Historical trend analysis and reporting

Web Dashboard: Responsive web interface offers:

- Comprehensive health analytics
- Multi-pet management capabilities
- Veterinary integration features
- Export functionality for health records

E. Emergency Response Implementation

The emergency response system provides comprehensive safety features:

Hardware SOS Implementation: Physical button triggers immediate response:

- GPIO interrupt for instant detection
- Buzzer activation for audio feedback
- LED indicator for status confirmation
- Immediate GPS location acquisition

Software Emergency Protocol:

- 1) SOS button detection and validation
- 2) High-priority GPS location request
- 3) Emergency SMS transmission with location
- 4) Cloud notification for mobile app alerts
- 5) Continuous tracking until acknowledgment

V. RESULTS AND DISCUSSION

A. System Performance Assessment

Comprehensive testing was conducted over 10 weeks using 12 dogs across different breeds, sizes, and activity levels to evaluate system performance, accuracy, and reliability.

Health Parameter Precision Metrics:

- Temperature Monitoring: $\pm 0.3^\circ\text{C}$ accuracy with 98.2% readings within clinical range
- Heart Rate Detection: 96.2% accuracy during rest, 93.4% during moderate activity
- Activity Classification: 94.7% overall accuracy (Resting: 96.1%, Walking: 93.8%, Running: 91.2%)

GPS and Location Performance:

- Outdoor Accuracy: 2.1-m average error in open environments
- Urban Environment: 4.3-m average error in urban areas
- Time to First Fix: Cold start 26 seconds, hot start 3.1 seconds

Cellular Communication Dependability:

- Data Transmission Success: 98.1% across multiple networks
- SMS Delivery: 99.3% delivery rate within 12-second average
- Network Coverage: Successful operation in 96% of tested locations
- Data Usage: Average 2.1 MB per month per device

Battery Performance and Power Management:

- Standard Operation: 6.1 days with 5-minute transmission intervals
- Power Saving Mode: 9.2 days with 15-minute intervals
- Emergency Mode: 48 hours continuous tracking

B. Comparative Assessment

Table I presents a comparison with existing commercial solutions:

TABLE I
COMPARISON WITH EXISTING SOLUTIONS

Feature	Our System	Fi Collar	Tractive
Communication	GSM/4G	LTE Cat-M1	2G/LTE
Monthly Cost	\$10	\$19	\$8–12
Battery Life	6+ days	3 months	2–5 days
Health Monitoring	Temp, HR, Activity	Activity	Activity
GPS Accuracy	2.1 m	3–5 m	2.5 m
Emergency Features	SOS + SMS	Location alerts	Safe zones
Weight	95 g	35 g	50 g
Water Resistance	IPX7	IPX68	IPX7

C. User Experience Assessment

Field testing with 18 dog owners provided comprehensive user experience feedback:

Setup and Installation:

- Average setup time: 15 minutes including SIM activation
- User satisfaction: 4.2/5.0 rating for ease of installation
- Technical support: 8% required assistance with cellular connection

Feature Usage:

- Health Monitoring: 93% regularly used, 82% found early health issue detection valuable
- GPS Tracking: 98% used location features, 91% successful pet recovery rate
- Emergency Alerts: 12 genuine emergencies with 100% successful alert delivery

D. System Constraints

Technical Limitations:

- Indoor GPS performance limited, mitigated by cellular triangulation
- Cellular coverage dependency affects remote area performance
- Battery life trade-off with comprehensive monitoring features

Deployment Considerations:

- PPG sensor performance affected by thick, dark fur
- Current design suitable for dogs > 8 kg
- Monthly cellular costs may limit adoption for price-sensitive users

VI. CONCLUSION AND FUTURE WORK

This research presented a comprehensive SIM-based smart collar system for real-time dog health monitoring and location tracking using cellular IoT technologies. The implementation successfully demonstrates significant improvements over existing solutions through autonomous cellular operation, integrated health monitoring, and extended battery life.

A. Primary Contributions

Our research contributes to pet monitoring technology through:

- 1) **Independent Cellular Framework:** Mobile network independence enabling global deployment with 98.1% transmission reliability
- 2) **Unified Multi-Parameter Monitoring:** Combined temperature, heart rate, activity classification, and GPS tracking with 96.2% overall accuracy
- 3) **Energy-Optimized Design:** 6+ days continuous operation through optimized power management and cellular protocols
- 4) **Complete Emergency Response:** SOS functionality with 99.3% SMS delivery reliability and automated alert distribution
- 5) **Economical Operation:** \$10/month cellular service cost with enhanced functionality compared to existing solutions

B. Future Research Opportunities

Advanced Health Analytics: Integration of machine learning algorithms for predictive health analysis and breed-specific health model development for early disease detection.

Sensor Technology Enhancement:

- Non-invasive blood glucose monitoring for diabetic pets
- Respiratory rate detection through chest movement analysis
- Stress level assessment using heart rate variability patterns

Communication Technology Evolution:

- 5G network integration for enhanced data transmission and reduced latency
- Satellite communication backup for remote area coverage

- Edge computing implementation for local AI processing

Multi-Pet Network Systems: Mesh networking capabilities enabling multiple pets to share health and location data for comprehensive household monitoring.

Veterinary Integration: Development of APIs for direct integration with veterinary practice management systems and telemedicine platforms.

The successful validation demonstrates the viability of cellular IoT technologies for autonomous pet monitoring, providing a foundation for next-generation animal health systems that could transform veterinary care and pet ownership through continuous, intelligent monitoring capabilities.

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